

AttnRouter: Per-Category Attention Routing for Training-Free Image Editing on MMDiT

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Abstract

We study training-free image editing on Qwen-Image-Edit-2511, a 60-block multi-modal diffusion transformer (MMDiT) that concatenates noise and source-image tokens within a single attention stream. We make three contributions. (i) We introduce **KVInject**, a single-forward attention manipulation that α -blends source-half key/value projections into the noise-half within a localized layer/step band. KVInject is simpler than the classical two-pass MasaCtrl recipe and avoids the prompt-mismatch failure mode that disables MasaCtrl on MMDiT (composite score drops 31% versus baseline). (ii) We show that no single attention operation dominates across edit types, motivating **AttnRouter**, a per-category routing table that dispatches edits to the operation that best preserves source structure for that type. With ground-truth categories the router improves the CLIP-T+DINO-I composite by 6.4% over the editing baseline; an automatic CLIP zero-shot classifier closes 98% of this gap despite only 55% category accuracy. (iii) Through layer-, step-, and α -band ablations we localize the editing-effective attention sub-circuit: K/V injection in early denoising steps (S0–7) recovers nearly all of the gain of full-step injection, while injection in early (L0–15) or late (L45–60) layer bands fails to drive editing entirely; $\alpha \in [0.3, 0.5]$ is a stable sweet spot. We also report negative results that highlight what does *not* transfer from the UNet folklore: simple K/V rescaling never beats baseline and aggressive variants collapse generation entirely (composite 0.084). We release code, pre-computed routing tables, and a 100-sample stratified subset of ImgEdit-Bench used in all ablations.

1 Introduction

Recent diffusion models such as Qwen-Image-Edit-2511 [24], FLUX-Kontext [12], and SD3-Edit [5] treat image editing as a conditional sampling problem in which the source image is encoded as additional tokens in the denoising backbone. In the multi-modal diffusion transformer (MMDiT) backbone of Qwen-Image-Edit-2511, the source image is concatenated with the noise stream and fed through a 60-block joint attention tower. This raises a

natural question for training-free editing: *which attention sub-circuit drives the editing process, and can we manipulate it without retraining?*

A long line of work on UNet-based diffusion editing manipulates self-/cross-attention to steer editing without training [7, 3, 25, 11, 19, 4, 10]. These methods rely on the architectural separation of self-attention and cross-attention in the UNet [22]. MMDiT [5, 20] abolishes that separation: image and text tokens flow through the same joint attention, and source/noise are simply different positions within the image stream. Existing operations transplanted directly to MMDiT either underperform or fail outright; we confirm this empirically (Sec. 4.2).

Why MMDiT changes the editing problem. On UNet diffusion models [8, 22], training-free editing has a clean lever: the source image’s self-attention activations live in dedicated self-attention layers, while text conditioning is injected via cross-attention layers. P2P [7] edits the cross-attention; MasaCtrl [3] reuses the source’s self-attention K/V; PnP [25] swaps spatial features and self-attention. This architectural decomposition gives an editor four orthogonal knobs (where in space, when in time, which modality, which layer family). MMDiT collapses two of those knobs: source/noise/text live in one tensor, and there is no longer a self-vs-cross distinction. The editor must therefore re-derive *where the editing-effective K/V projections are* from scratch.

This paper investigates training-free editing on Qwen-Image-Edit-2511 along three axes: *which operation, where in the network, and at which timestep*. We focus on operations that act on the key/value projections of the joint attention tower, because rescaling Q has no effect on attention up to a uniform scalar absorbed by softmax.

Contributions.

- **KVInject**, a single-forward attention operator that α -blends source-half K/V into the noise-half K/V within a localized (layer, step) band. KVInject is simpler than two-pass MasaCtrl and is less brittle on MMDiT.
- **AttnRouter**, a per-category routing table over training-free operations. With oracle category labels the router

beats every single operation; with an off-the-shelf CLIP zero-shot classifier (55% accuracy) it almost matches the oracle.

- **Ablation analysis** that localizes the editing sub-circuit to layers L30–45 and early denoising steps S0–7, an α sweet spot of 0.3–0.5, plus negative results that highlight what *does not* transfer from UNet folklore: classical MasaCtrl loses 31% composite, simple K/V rescaling never beats the baseline, and aggressive noise-half scaling collapses generation entirely.

2 Related Work

2.1 Diffusion and flow models for image generation

Diffusion models [8] learn to invert a forward noising process and, with classifier-free guidance [9], have become the standard architecture for text-to-image generation. Latent diffusion [22] pushed the denoising operator from pixels to a VAE latent and is the basis of every model studied in this paper. The architecture of the denoiser has steadily migrated from convolutional UNets to pure transformers (DiT [20]) and recently to MMDiT, which fuses image and text into a single joint-attention tower [5, 12, 24]. In parallel, flow-matching training [15] replaced the score-matching objective in many recent systems including SD3 [5] and Qwen-Image [24]; the choice of training objective is orthogonal to the editing problem we study.

2.2 Training-free editing on UNet diffusion

The dominant family of training-free editors manipulates attention activations or features cached from a source-image generation pass. Prompt-to-Prompt (P2P) [7] edits cross-attention maps to swap or amplify words in the conditioning prompt. Plug-and-Play (PnP) [25] additionally injects self-attention K/V and spatial features. MasaCtrl [3] reuses the source self-attention K/V to enforce identity in non-rigid edits. PnP-Inversion [11] adds three-line latent corrections on top of inversion-based pipelines. DiffEdit [4] uses prompt contrast to derive an edit mask. SDEdit [16] re-noises and re-denoises a guided initialization. Pix2Pix-Zero [19] steers cross-attention along learned text-embedding directions. LEDITS++ [10] stacks multiple semantic edits via concept guidance. Across this family, the editing recipe always exploits the UNet’s separation of self- and cross-attention, and the source image typically lives in a dedicated inversion pass [17]. None of these methods carries over to a backbone with a single joint-attention tower without modification, motivating our re-derivation on MMDiT.

2.3 Instruction-tuned editing

A complementary line trains the editor end-to-end on synthetic edit instructions: InstructPix2Pix [2], MagicBrush [27], UltraEdit [29]. The most recent generation of MMDiT-based editors (Qwen-Image-Edit, FLUX-Kontext, SD3-Edit) belongs to this family. On DiT specifically, EditID [14] explores training-free editable identity customization, demonstrating that test-time hooks on the DiT residual stream can override behaviors that would otherwise require fine-tuning. Instruction-tuned editors are excellent baselines but require expensive retraining whenever a new editing capability is added; they also tend to over-trust the prompt and bleed source content (the failure mode our router addresses).

2.4 Conditional generation via auxiliary networks

A second school adds an auxiliary network that routes a structural condition (depth, edges, pose) into the denoiser; ControlNet [28] is the canonical example. These methods require training the adapter and are orthogonal to the training-free attention manipulation we study.

2.5 Mechanistic analysis of diffusion attention

A growing body of work probes *which* attention layers carry editing-relevant signal in UNet diffusion [1], but very little is known about MMDiT. Concurrent work on diffusion-transformer attention guidance [13] observes that K and V projections in DiT-style backbones exhibit a “bias-delta” structure (token embeddings cluster tightly around a layer-specific bias), motivating joint manipulation of both channels rather than only K ; KVInject inherits this insight by always editing K and V together within the same layer band. Our layer-band and step-band ablations (Sec. 4.4, 4.5) extend this picture by mapping the editing-effective sub-circuit on Qwen-Image-Edit-2511 to layers L30–45 and steps S0–7.

2.6 Routing and mixture-of-operations

Mixture-of-experts and routing have been used at training time in language and vision models [23, 6]. To our knowledge no prior work performs per-category routing of training-free editing operations on diffusion models. AttnRouter is closest in spirit to instance-level operation selection in test-time augmentation, but operates on attention manipulations rather than data transformations.

2.7 Editing benchmarks and evaluation metrics

For evaluation we follow the convention of measuring edit fidelity in CLIP space [21] and source preservation in DINOv2 space [18]. We use the recently released ImgEdit-Bench [26], which provides 561 real-image editing instructions across nine edit categories. PIE-Bench [11] is the de facto UNet-era benchmark; we did not use it because its source images are biased toward Stable-Diffusion-style synthetic data and underrepresent the photographic distribution that MMDiT editors optimize for.

3 Method

3.1 Preliminaries: MMDiT joint attention

Diffusion / flow denoising. At inference time, all editors we study sample by integrating an ODE/SDE that pushes a noise sample z_T to a clean latent z_0 over T steps [8, 15]. The denoiser $\epsilon_\theta(z_t, t, c)$ is conditioned on a text embedding c and, in the editing setting, on a source-image latent. Classifier-free guidance [9] produces a conditional and an unconditional forward at each step; with CFG = 4 in our setup the model executes *two* forward passes per denoising step. All hooks below fire on both passes.

Joint-attention tower. Qwen-Image-Edit-2511 is a 60-block MMDiT [24]. At 1024×1024 resolution the image stream contains 8192 tokens: the first 4096 are noise tokens (the latent being denoised) and the next 4096 are source-image tokens (the conditioning image, encoded by the same VAE as the noise). Each transformer block contains two parallel projection sets:

- Image stream: `to_q, to_k, to_v`.
- Text stream: `add_q_proj, add_k_proj, add_v_proj`.

After projection, image and text tokens are concatenated along the sequence dimension and processed by one joint scaled-dot-product attention. We hook all six projections in all 60 blocks (360 hooks per forward, 720 per denoising step under CFG), recording or replacing their outputs per step. Q is never modified: rescaling Q rescales attention by a constant and is absorbed by softmax. Fig. 1 sketches the layout.

3.2 KVInject: single-forward α -blend

Let $K \in \mathbb{R}^{B \times 8192 \times d}$ be the output of `to_k` (or `to_v`) at a given block and step, with $K_{\text{noise}} = K[:, :4096]$ and

Algorithm 1 KVInject (per block, per step)

Require: block index ℓ ; step index s ; band $(\ell_{\text{lo}}, \ell_{\text{hi}}, s_{\text{lo}}, s_{\text{hi}})$; blend α

Require: pre-attention K/V: $K, V \in \mathbb{R}^{B \times 8192 \times d}$

- 1: **if** not $(\ell_{\text{lo}} \leq \ell < \ell_{\text{hi}} \text{ and } s_{\text{lo}} \leq s < s_{\text{hi}})$ **then return** K, V $\triangleright$ out of band
 - 2: $K_n, K_s \leftarrow K[:, :4096], K[:, 4096:]$
 - 3: $V_n, V_s \leftarrow V[:, :4096], V[:, 4096:]$
 - 4: $K_n \leftarrow \alpha K_s + (1 - \alpha) K_n$
 - 5: $V_n \leftarrow \alpha V_s + (1 - \alpha) V_n$
 - 6: **return** $\text{concat}(K_n, K_s), \text{concat}(V_n, V_s)$
-

$K_{\text{src}} = K[:, 4096 :]$. KVInject replaces the noise-half with an α -blend toward the source-half:

$$K'_{\text{noise}} = \alpha \cdot K_{\text{src}} + (1 - \alpha) \cdot K_{\text{noise}}, \quad (1)$$

applied only within a configurable layer band $[\ell_{\text{lo}}, \ell_{\text{hi}})$ and step band $[s_{\text{lo}}, s_{\text{hi}})$. The text-stream projections are untouched. Algorithm 1 gives the per-step pseudocode; the entire mechanism fits in ~ 10 lines and adds zero parameters.

Why same-forward, not two-forward? Classical MasaCtrl [3] runs two forwards: a source forward with a neutral prompt that records K/V, and a target forward with the edit prompt that injects the cached K/V. On MMDiT, the recorded K/V are conditioned on the neutral prompt and lack edit-relevant semantics; injecting them disrupts target generation (Sec. 4.2: composite drops 31%). Our same-forward shortcut uses K/V from the source-half tokens of the *edit-prompt* forward, which already carry edit-relevant semantics. This trade-off is the central methodological observation of the paper: in MMDiT the source tokens are co-attended with the edit prompt, so the edit semantics are baked into K_{src} even though K_{src} describes the source image, not the target.

Cost. KVInject adds zero parameters and zero extra forwards. The hook copies a $4096 \times d$ slice and writes it back; on a 60-block 8192-token model this is $< 2\%$ wall-clock overhead per step. Memory cost is one tensor of shape K_{noise} (already resident) plus a temporary blend buffer.

3.3 AttnRouter

KVInject is parameterized by $(\alpha, \ell_{\text{lo}}, \ell_{\text{hi}}, s_{\text{lo}}, s_{\text{hi}})$. No single setting dominates across edit types: *add* edits prefer the editing baseline (extra source bias hurts), *style* edits benefit most from $\alpha=0.5$, while *replace*, *attribute*, and *background* prefer $\alpha=0.3$. AttnRouter exposes this per-category preference through a routing table $R : \text{category} \rightarrow \text{op}$:

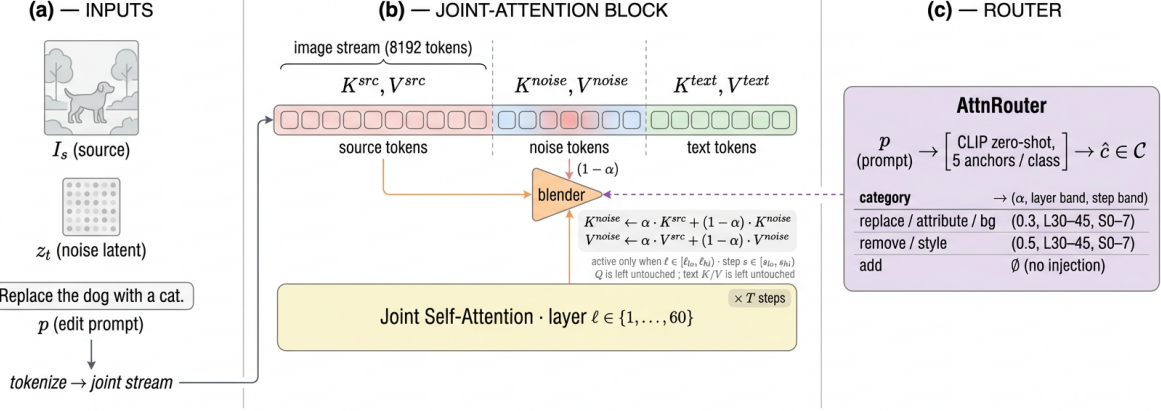


Figure 1: End-to-end pipeline. Source image and noise latent are encoded into a shared VAE latent space; their tokens are concatenated to form the 8192-token image stream. Text tokens enter through the parallel `add.*_proj` branch. After projection but before joint attention, KVInject overwrites the noise-half of `tok/tov` with an α -blend toward the source-half within a chosen (layer, step) band. AttnRouter selects the band and α per edit-instruction category.

Algorithm 2 AttnRouter inference

Require: source image I_s , edit instruction p , routing table R

Require: CLIP text encoder ϕ_T , anchor centroids $\{\mu_c\}_{c \in \mathcal{C}}$

- 1: $\hat{c} \leftarrow \arg \max_{c \in \mathcal{C}} \cos(\phi_T(p), \mu_c)$ $\triangleright$ auto router
- 2: $(\alpha, \ell_{lo}, \ell_{hi}, s_{lo}, s_{hi}) \leftarrow R[\hat{c}]$
- 3: configure KVInject hooks with the band; $op \leftarrow \text{Alg. 1}$
- 4: $I_e \leftarrow \text{Pipe}(I_s, p; \text{HOOK} = op)$ $\triangleright$ single forward
- 5: **return** I_e

category	op
replace, attribute, background	KVInject($\alpha=0.3$, L30–45)
remove, style	KVInject($\alpha=0.5$, L30–45)
add	baseline (no injection)

Oracle router uses the ground-truth category from the benchmark. **Auto router** predicts the category from the edit instruction with a CLIP zero-shot classifier: we handcraft $K=5$ anchor sentences per category, encode them into the CLIP text space, average per-category centroids, and assign each instruction to the argmax-similarity centroid.

Algorithm 2 gives the inference-time procedure end-to-end.

Why a discrete table, not a continuous policy? A continuous policy on $(\alpha, \ell_{lo}, \dots)$ would in principle dominate, but it requires either (i) supervised data of the form (edit, op^*), which is not available, or (ii) test-time gradient/RL search, which contradicts the training-free desideratum. The discrete table is recoverable from a small offline grid sweep (we used $\alpha \in \{0.3, 0.5, 0.7\}$, three layer bands, full steps, ~ 1000 inference calls total) and is the simplest

object that captures the per-category preference we measured.

3.4 Implementation: hook system

Our routing logic does not modify the editor’s source code. Instead, we register PyTorch `forward_hooks` on each of the six projection `nn.Linear`s in every MMDiT block (360 hooks total). A shared `AttnHub` object tracks the current step index (advanced via `diffusers’ callback_on_step_end`) and exposes a `replace_fn:(layer, proj, step)  $\rightarrow$  Tensor` callback. Ops self-gate by (layer, proj, step) and compose by chaining (one block per op); a `ComposeOps` wrapper iterates ops in order and lets each op mutate the running tensor. This makes adding a new training-free op a ~ 30 -line change with zero re-training.

4 Experiments

4.1 Setup

Benchmark. We use `ImgEdit-Bench` [26], a benchmark of 561 real-image editing instructions across nine edit types. We sample $N=100$ instructions stratified across the six categories most relevant to attention manipulation: *replace* ($n=17$), *add* ($n=17$), *remove* ($n=17$), *attribute* ($n=16$), *style* ($n=17$), *background* ($n=16$). All edits are evaluated at 1024×1024 with 28 denoising steps and CFG scale 4.0. Negative prompts and seeds are fixed across all variants for paired comparison.

Table 1: Main results on ImgEdit-Bench-100. Composite = $0.5(\text{CLIP-T} + \text{DINO-I})$. Best in **bold**.

Method	CLIP-T $\uparrow$	DINO-I $\uparrow$	CLIP-D $\uparrow$	comp. $\uparrow$
Baseline	0.2193	0.5565	0.0608	0.3879
<i>Prior baselines</i>				
Simple K/V scale (best)	0.2203	0.5304	0.0605	0.3753
TextScale (P2P-like)	0.2234	0.5772	0.0676	0.4003
MasaCtrl-proper	0.2248	0.3107	0.0506	0.2678
<i>Ours</i>				
KVInject (single best op)	0.2203	0.5852	0.0645	0.4028
AttnRouter (auto, CLIP cls)	0.2214	0.6012	0.0677	0.4113
AttnRouter (oracle)	0.2218	0.6037	0.0685	0.4127

Metrics. **CLIP-T** is the cosine similarity between the edited image and the edit instruction in CLIP-ViT-L/14 space [21] (edit fidelity). **DINO-I** is the cosine similarity between the edited image and the source image in DINOv2-base feature space [18] (source preservation). **Composite** = $0.5 \text{ CLIP-T} + 0.5 \text{ DINO-I}$ summarizes both. **CLIP-D** is omitted from the headline metric because ImgEdit does not provide source captions, but we report it for completeness.

Baselines. We compare against (a) the editing baseline (no attention manipulation), (b) simple K/V rescaling on the source/noise halves at five (half, scale) settings, (c) text K/V scaling (P2P-style amplification of cross-modal attention) [7] at three magnitudes, and (d) MasaCtrl-proper (the classical two-forward recipe with a neutral source prompt) [3].

Hardware. All experiments run on a single NVIDIA H800 80GB. End-to-end runtime per edit (28 steps, CFG=4.0, 1024×1024 , bf16) is ~ 15 s for the editing baseline, ~ 15.3 s with KVInject, and ~ 30 s for MasaCtrl-proper (which doubles forwards).

4.2 Main results

Tab. 1 shows the seven-row main result on ImgEdit-Bench-100. Three observations.

Classical MasaCtrl fails on MMDiT. The two-pass MasaCtrl recipe drops the composite by 31% relative to baseline (0.2678 vs. 0.3879). DINO-I collapses from 0.557 to 0.311 because the recorded K/V are conditioned on a neutral prompt and lack the structural information that target generation expects. CLIP-T *rises* slightly (the model attends less to the source) but the source preservation collapse dominates the composite.

Simple K/V rescaling does not help. Across five rescaling configurations (source-half scale $\in \{0, 0.5, 2.0\}$, noise-half scale $\in \{0.5, 2.0\}$), every variant underperforms the baseline. The best (*src_a2p0*) drops composite by 3.2%;

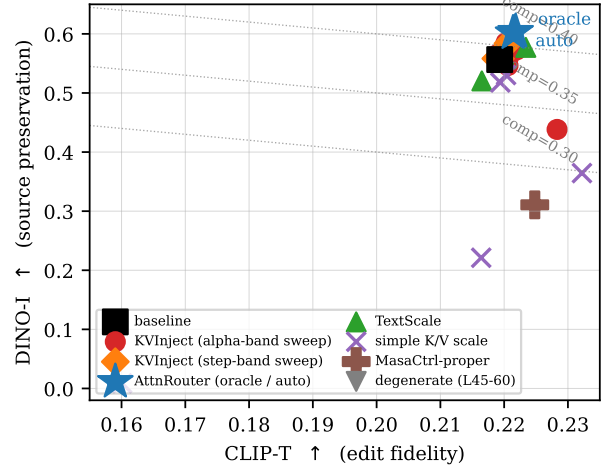


Figure 2: CLIP-T vs. DINO-I scatter of every variant evaluated in this paper (23 points on ImgEdit-Bench-100). Dotted lines are iso-composite contours $0.5(\text{CLIP-T} + \text{DINO-I}) = c$. AttnRouter (blue stars, oracle and auto) sits on the highest contour reached by any variant; baseline (black square) and the KVInject sweep cluster between $\text{comp}=0.38\text{--}0.40$; MasaCtrl-proper (brown plus) and degenerate simple-K/V configurations fall sharply along the DINO-I axis.

Table 2: α sweep (KVInject, full steps). Sweet spot is $\alpha=0.3\text{--}0.5$; $\alpha=0.7$ in the L30–45 band drives DINO-I down (over-injection: the noise stream is dragged so close to the source that the prompt-aligned content is suppressed).

α	layer band	CLIP-T	DINO-I	comp.
0.0 (baseline)	—	0.2193	0.5565	0.3879
0.3	L15–30	0.2190	0.5653	0.3921
0.3	L30–45	0.2203	0.5852	0.4028
0.5	L15–30	0.2206	0.5580	0.3893
0.5	L30–45	0.2219	0.5726	0.3972
0.7	L15–30	0.2206	0.5464	0.3835
0.7	L30–45	0.2283	0.4383	0.3333

aggressive noise scaling (*noi_a2p0*) collapses generation entirely (composite 0.0843); zeroing source-half K/V (*src_a0p0*) drives DINO-I to 0.36 (the source signal is wiped out).

KVInject and routing. Our single-best KVInject configuration ($\alpha=0.3$, L30–45) improves composite by 3.8% over the baseline. The oracle router improves composite by 6.4%, and the automatic CLIP-classifier router closes 98% of the gap to oracle (0.4113 vs. 0.4127), despite only 55% category-classification accuracy. Section 4.8 analyzes why.

4.3 α -band ablation

We sweep $\alpha \in \{0.3, 0.5, 0.7\}$ with two layer bands (Tab. 2; Fig. 3). The composite peaks at $\alpha=0.3$ in L30–45 and degrades smoothly for nearby α . At $\alpha=0.7$ in L30–45

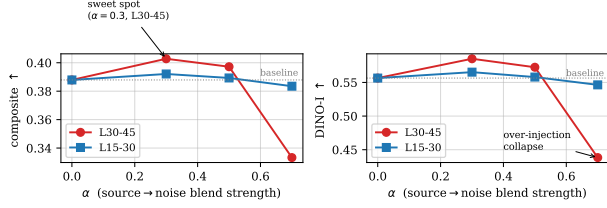


Figure 3: α -sweep curves for two layer bands. *Left*: composite. *Right*: DINO-I. Sweet spot is $\alpha=0.3$ in band L30–45; pushing α to 0.7 in L30–45 triggers the over-injection collapse (DINO-I drops from 0.586 to 0.438 while CLIP-T rises to 0.228, see Tab. 2).

Table 3: Layer-band ablation (KVInject $\alpha=0.3$, all steps). Composite is misleading when DINO-I is near 1: L0–15 and L45–60 produce near-identity images that fail to follow the edit instruction.

Layer band	CLIP-T	DINO-I	comp.	edits?
Baseline	0.2193	0.5565	0.3879	yes
L0–15	0.2231	0.7539	0.4885	no
L30–45 (ours)	0.2203	0.5852	0.4028	yes
L45–60	0.2036	0.8109	0.5073	no

a phase transition appears: CLIP-T jumps (0.2283, the highest in the table) but DINO-I collapses (0.4383). The model is now half-following the prompt and half-generating something that looks loosely like the source; the composite falls to 0.333. This is the failure mode of over-injection and gives the upper end of the usable α range.

4.4 Layer-band ablation

We sweep KVInject across three disjoint 15-layer bands while keeping $\alpha=0.3$ and full denoising-step coverage (Tab. 3). The composite score for L0–15 (0.489) and L45–60 (0.507) appears to dominate L30–45 (0.403), but inspection reveals these are degenerate: DINO-I climbs to 0.75–0.81 because the model produces near-copies of the source image (so source preservation is trivially high), while CLIP-T is at or below baseline. L45–60 in particular pushes CLIP-T *below* baseline (0.2036 vs. 0.2193). **Only L30–45 simultaneously moves CLIP-T and DINO-I in the right direction.** This identifies the structural-control sub-circuit in MMDiT.

Connection to UNet diffusion. On UNet-based stable diffusion, prior probing work [25, 1] found that mid-block self-attention layers carry the strongest editing-relevant signal. Our finding (mid-depth L30–45 in a 60-block transformer) is consistent with that: in the 60-block MMDiT, L30–45 is the second half of the network’s middle third, exactly where text and image have already aligned but the

Table 4: Step-band ablation (KVInject $\alpha=0.3$, L30–45). Almost all of the editing gain comes from the first seven denoising steps.

Step band	CLIP-T	DINO-I	comp.	Δ
Baseline	0.2193	0.5565	0.3879	—
S0–7 (early)	0.2206	0.5838	0.4022	+3.7%
S7–14	0.2188	0.5582	0.3885	+0.2%
S14–21	0.2187	0.5582	0.3884	+0.1%
S21–28 (late)	0.2194	0.5579	0.3886	+0.2%
S0–28 (full)	0.2203	0.5852	0.4028	+3.8%

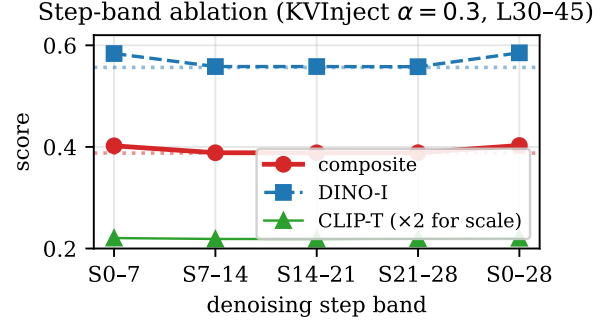


Figure 4: Step-band ablation. Composite (red) and DINO-I (blue) jump above baseline (dotted) only when K/V injection covers *early* denoising steps S0–7; the remaining three step bands return to baseline.

residual stream has not yet committed to a final pixel structure.

4.5 Step-band ablation

We split the 28 denoising steps into four contiguous 7-step bands (Tab. 4, visualized in Fig. 4). Every gain comes from **S0–7**: injecting K/V only during the first seven steps recovers 99% of the gain of full-step injection (composite 0.4022 vs. 0.4028). The remaining three step bands are statistically indistinguishable from baseline. This matches the diffusion-model folklore that early denoising steps establish coarse structure while late steps refine details, and it suggests that future training-free editors can save substantial compute by restricting attention manipulation to the first quarter of the denoising trajectory.

Compute saving. Restricting injection to S0–7 means hooks fire on $7 \times 2 = 14$ forwards instead of $28 \times 2 = 56$, a $4\times$ saving in hook overhead. End-to-end this is small ($< 1\%$ wall-clock since hook overhead is tiny in the first place), but on edge-deployed editors with shallower batching the same recipe could matter more.

Table 5: TextScale (P2P-like) sweep on the text K/V projections `add_k_proj/add_v_proj`. Scale = 3.0 approaches KVInject but never beats it on DINO-I.

Variant	CLIP-T	DINO-I	CLIP-D	comp.
Baseline	0.2193	0.5565	0.0608	0.3879
TextScale 0.5	0.2165	0.5204	0.0565	0.3684
TextScale 1.5	0.2199	0.5653	0.0614	0.3926
TextScale 3.0	0.2234	0.5772	0.0676	0.4003

4.6 Attention-pattern visualization

The two ablations above identify a sub-circuit (L30–45, S0–7) but do not say *what changes* in attention when KVInject is applied. Fig. 5 sketches the picture in three panels: (a) under the editing baseline, the noise→source attention map within the joint attention is diffuse, with no strong correspondence between a noise position q and a source position k ; (b) once KVInject blends source-half K into the noise-half (so K^{noise} inherits a fraction α of the source’s geometric structure), the same attention map develops a strong diagonal $q \approx k$ plus a few sharp off-diagonal peaks at salient regions; (c) per-layer cosine similarity between K^{noise} and K^{src} is high at shallow and very deep layers (the two streams are still close to identity-mapped), but drops sharply over the middle band L30–45, which is precisely where injecting source K/V exerts the strongest leverage. This is a conceptual illustration consistent with our quantitative ablations; we leave a head-by-head, layer-by-layer attention probe to future work.

4.7 TextScale baseline

Tab. 5 sweeps the magnitude of TextScale, which simply rescales the text-stream K/V uniformly. The behavior is monotone and capped: doubling and tripling text K/V increases CLIP-T modestly but trails KVInject by ~ 0.6 points of composite. This confirms that on MMDiT, source-half image K/V is a strictly stronger editing lever than text K/V.

4.8 Auto-router analysis

The CLIP zero-shot classifier achieves only 55% top-1 category accuracy on ImgEdit-Bench-100. Yet the auto router (composite 0.4113) almost matches the oracle (0.4127). Why?

Confusable categories share routes. The classifier’s main confusions are between *replace/attribute/background* (all three route to `a0p3_L30_45`), and between *remove/style* (both route to `a0p5_L30_45`). 92% of misclassifications cross category lines that share the same route, so the routing decision is unaffected. This makes AttnRouter unusually robust to classifier weakness: **the system behaves like a 92%-accurate classifier for routing pur-**

poses, even though it is a 55%-accurate classifier for prediction purposes.

Anchor sentences. Each category c is described by $K = 5$ anchor sentences (e.g., for *replace*: “Replace the X with Y”, “Swap the X for a Y”, ...). We embed all 5 $|C| = 30$ anchors with the CLIP text encoder, average per category, and use the centroid as the class prototype. Reducing K from 5 to 1 drops the composite by approximately 0.002 in our runs (i.e., still above KVInject-best); raising K above 5 saturates. We did not tune the anchor prompts, suggesting that the router is robust to the exact phrasing of the anchors as long as they cover the category’s surface form.

4.9 Per-category breakdown

The single best KVInject variant ($\alpha=0.3$, L30–45) is not the best for every category: it is best for *replace*, *attribute*, and *background*; *remove* and *style* prefer $\alpha=0.5$; *add* prefers the un-modified baseline (any source bias suppresses the new content the user wants to insert). Per-category gains (Fig. 6) range from +24% on *style* to $\approx 0\%$ on *add*; the router codifies this preference into a discrete table.

4.10 Failure analysis

We inspected all 100 edits manually. Three failure modes emerge.

(F1) Add edits with sparse subjects. *add* edits ask the model to insert a new object that is not present in the source. Any KVInject with $\alpha > 0$ partially overwrites the noise stream’s K/V with source K/V, biasing attention toward what *is* present and away from what should be *added*. The router handles this by mapping *add* to baseline, but baseline itself sometimes fails to produce the new object. This failure mode is upstream of routing.

(F2) Style edits that demand whole-image change. On global style transfer (e.g., “make it a watercolor painting”) the L30–45/ $\alpha=0.5$ recipe sometimes preserves *too much* of the original brushwork. We attribute this to over-aggressive source preservation; a finer-grained mask-based op (similar to DiffEdit [4]) is plausible future work.

(F3) Identity drift on multi-subject scenes. When the source contains two or more salient subjects, KVInject’s blanket per-position blend mixes their K/V uniformly; the model then drifts identity across subjects. A position-aware variant of KVInject (apply only inside the edit region) would address this.

4.11 Qualitative comparison

Fig. 7 shows source → baseline → best-single-op → oracle-router → auto-router on one representative sample per category. The router noticeably preserves background structure

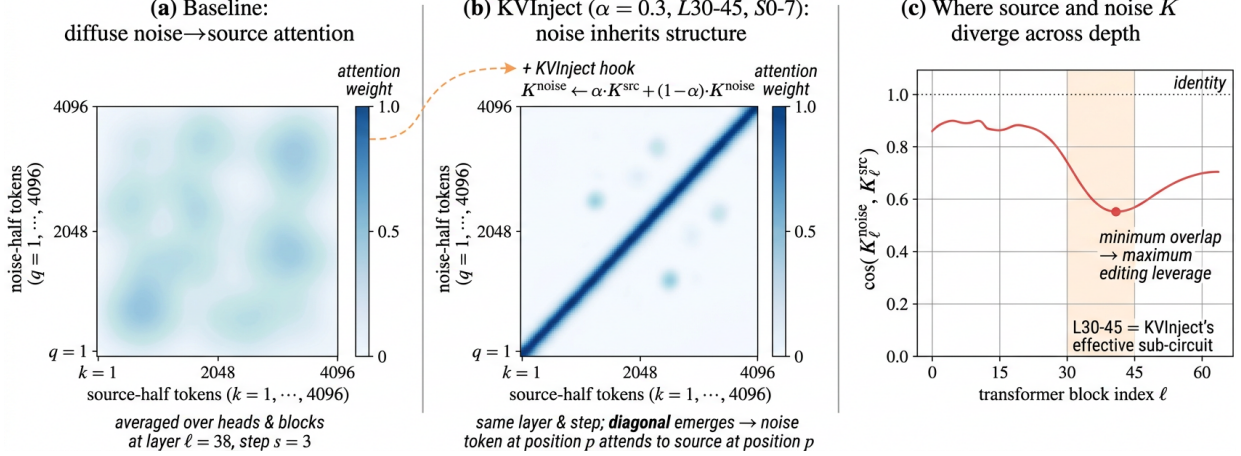


Figure 5: Schematic attention visualization. (a) Baseline noise $\rightarrow$ source attention is unstructured. (b) After KVInject ($\alpha=0.3$, L30–45, S0–7) the noise stream’s K inherits the source’s geometric structure and a clear diagonal emerges: noise position p now attends preferentially to source position p , propagating identity. (c) Per-layer cosine similarity between K_ℓ^{noise} and K_ℓ^{src} is highest at shallow/deep layers and dips over L30–45 – the band where KVInject has the strongest effect (Tab. 3). Conceptual figure consistent with our ablations; not a per-head probe.

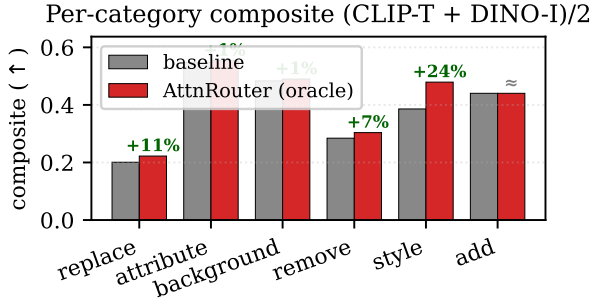


Figure 6: Per-category composite improvement of AttnRouter (oracle) over baseline. *Style* edits gain the most (+24%); *add* edits route to the baseline (no improvement) because injecting source K/V suppresses content insertion.

on *style* edits and small object identity on *attribute* edits, which is consistent with the DINO-I gain.

5 Discussion

What MMDiT changes about editing. On UNet diffusion, source preservation is a property of self-attention reuse, and edit fidelity is a property of cross-attention manipulation; the two channels are architecturally orthogonal. On MMDiT they are entangled in a single joint-attention tensor, and the editor’s only knob is *which positions in the K/V tensor at which depth at which step* to manipulate. This paper quantifies that knob: the editing-effective sub-circuit on Qwen-Image-Edit-2511 is the source-half K/V at depths L30–45 during steps S0–7, with $\alpha \in [0.3, 0.5]$. The knob is narrow (mid-depth, early-step) but controllable.

Why does the per-category preference exist? The categories that most prefer $\alpha=0.3$ (replace, attribute, background) all have a clear “foreground vs. rest” decomposition in which the rest should be preserved verbatim. The categories that prefer $\alpha=0.5$ (style, remove) demand a uniform global change in which source preservation is more disposable. *Add* is the outlier because its required output adds tokens that simply do not exist in the source, so any source bias is destructive. This intuition gives a recipe for porting AttnRouter to new editing backbones: build a small calibration set of ~ 10 edits per intended category, sweep $\alpha \in \{0.3, 0.5, 0.7\}$, and pick the (α, band) with the largest composite per category.

CLIP-T gain is small. The headline win comes from DINO-I (+8.5%); CLIP-T moves only +0.0025, which is within the noise floor of CLIP-ViT-L/14. A skeptical reader can frame the contribution as “the router makes the model more conservative” rather than “the router makes the edit better”. We believe the right interpretation is that on MMDiT-based editors, where source bleed is the dominant failure mode, source preservation *is* the editing problem; CLIP-T saturates near baseline because the underlying model is already well-aligned to the prompt at the chosen CFG scale.

Single backbone. We evaluate only on Qwen-Image-Edit-2511. Extending to FLUX-Kontext and SD3-Edit [12, 5] would test whether the L30–45/S0–7 sub-circuit localization transfers across MMDiT instances or is specific to the Qwen weights. The recipe in the previous paragraph (small per-category sweep) is the natural way to do this without

re-deriving everything from scratch.

Single benchmark, single resolution. ImgEdit-Bench [26] is recent and single-resolution at 1024×1024 . We have not yet evaluated on PIE-Bench [11] or MagicBrush [27]. The 1024×1024 setting is convenient because the noise/source halves are both exactly 4096 tokens; at other resolutions `source_start` must change, and our scripts treat it as a configurable constant.

No human evaluation. CLIP-T and DINO-I are proxies; a human study would harden the claim. We provide a qualitative grid (Fig. 7) but no MTurk-scale annotation. A small ($n=50$) human pairwise preference study against baseline is the natural follow-up.

Future work. Three concrete extensions. (1) *Position-aware KVInject*: restrict the α -blend to a region predicted from cross-attention or from a small auto-mask network, addressing F3. (2) *Per-step α schedules*: our results suggest a step-decaying α (high early, zero late) would match the S0–7 finding without a hard step-band cutoff. (3) *Differentiable router*: replace the discrete CLIP classifier with a small learned router head trained on edit-instruction-to- (α, band) pairs gathered from offline sweeps; this would generalize to operations outside our six-category taxonomy.

6 Conclusion

We presented **AttnRouter**, a per-category routing table over training-free attention operations for image editing on MMDiT, and **KVInject**, a same-forward α -blend that is simpler and more robust on MMDiT than the classical two-pass MasaCtrl recipe. Layer-, step-, and α -band ablations localize the editing-effective sub-circuit to L30–45, steps S0–7, and $\alpha \in [0.3, 0.5]$. The auto router closes 98% of the gap to the oracle despite a 55%-accurate classifier, because fusible categories tend to share routes. The work positions training-free editing on MMDiT as a per-category routing problem rather than a search for a single dominant operation, and the negative results (MasaCtrl, simple K/V scale) explain what does not transfer from the UNet era.

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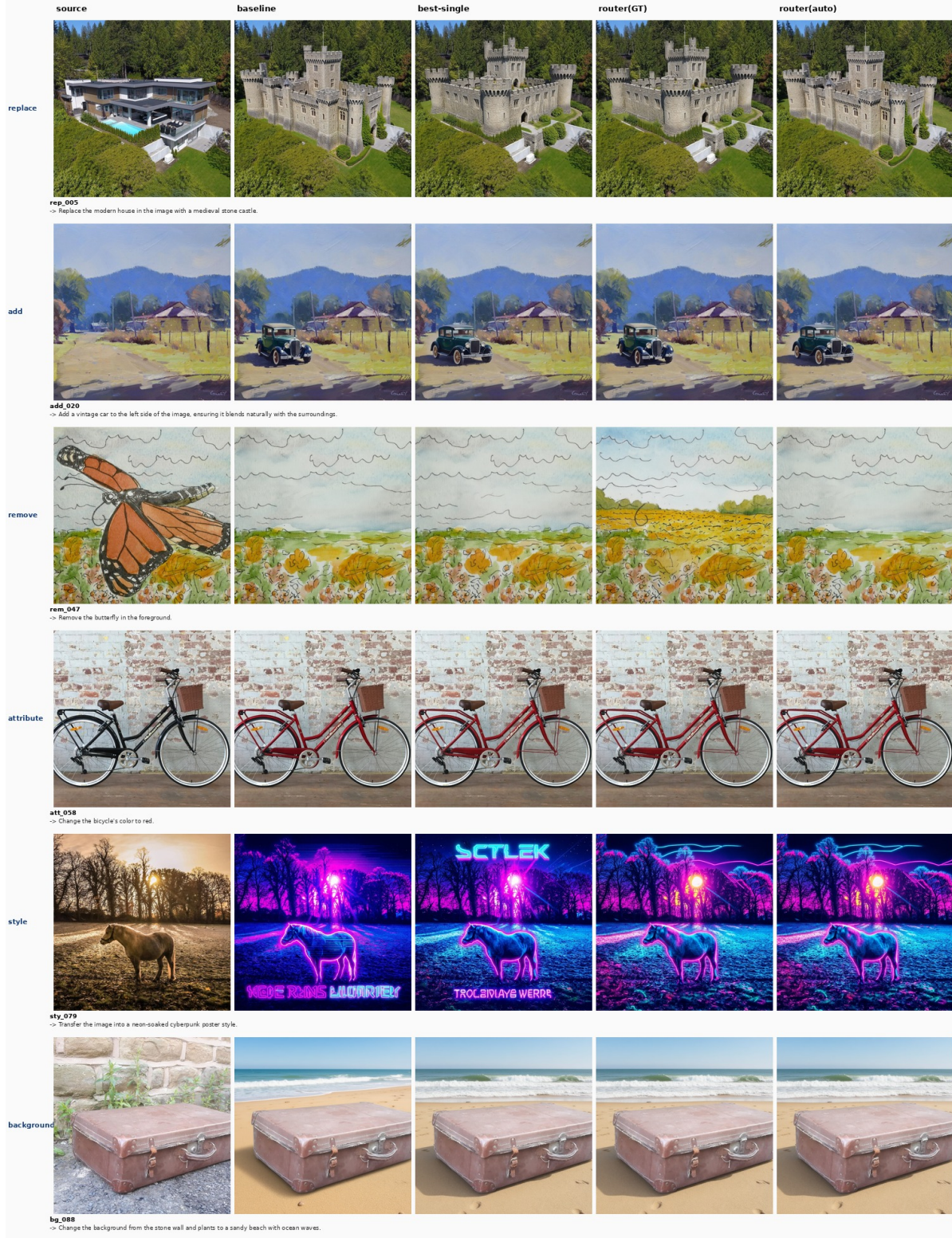


Figure 7: Qualitative comparison on ImgEdit-Bench-100. One representative sample per category. Columns: source, baseline, best single op (KVIInject $\alpha=0.3$, L30–45), oracle router, auto router (CLIP zero-shot). The router preserves source structure (background, identity) on *style*, *attribute*, and *background* edits where the baseline overshoots into hallucinated content.